Paleomagnetism of the Notre Dame Bay lamprophyre dikes, Newfoundland, and the opening of the North Atlantic Ocean[1]

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Lamprophyre dikes and the Leading Tickles dunite from Notre Dame Bay, Nfld., were sampled for a paleomagnetic study to define their age relationships and the tectonic implications of their emplacements. A single normal polarity direction (328°, +62°) was obtained by alternating field and thermal treatment, from 10 Cretaceous lamprophyre dikes. The paleopole (212°E, 67°N) derived from these dikes agrees with the cluster of North American Cretaceous paleopoles. On the other hand, the Leading Tickles dunite recorded two magnetizations: one (206°, +32°) related to the time of intrusion and interpreted to be Ordovician; the second one (000°, +66°), an overprint, is related to the intrusion of the lamprophyre dikes and interpreted to be Cretaceous. The polarity, paleopole, radiometric age, and tectonic setting of the lamprophyre dikes when compared with the Isachsen dikes, N.W.T., suggest the opening of the North Atlantic Ocean was initiated in the Lower Cretaceous.

Les filons de lamprophyre et l'intrusion de Leading Tickles, de la région de la Baie de Notre-Dame à Terre-Neuve, ont été échantillonnés afin de définir l'âge relatif ainsi que l'implication tectonique de l'emplacement de ces deux intrusions. Une direction unique de polarité normale (328°, +62°) a été obtenue, par champ alternatif et par traitement thermique, de 10 filons de lamprophyre d'âge Crétacé. Le pôle (212°E, 67°N) obtenu de ces filons correspond à l'essaim de pôles Crétacé relatif à l'Amérique du Nord. De l'intrusion de Leading Tickles, on a pu définir deux aimantations: la première (206°, +32°) relative à l'intrusion elle-même, est datée de l'Ordovicien; la deuxième (000°, +66°), une suraimantation, est reliée à l'intrusion des filons et est interprétée comme étant d'âge Crétacé.

La polarité, le paléopôle, l'âge radiométrique ainsi que l'emplacement tectonique des filons de lamprophyre ont été comparés aux filons d'Isachsen, T.N.O.; cette comparaison a permis de conclure que l'ouverture de l'océan Atlantique Nord a débuté au Crétacé inferieur.

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Introduction

The Central Mobile Belt of Newfoundland consists of oceanic crust overlain by volcanic and sedimentary rocks of Lower Paleozoic age (Williams *et al.* 1974; Williams and Stevens 1974, Fig. 1). Located in its northeastern end, the Notre Dame Bay area (Figs. 1, 2) comprises a sequence of volcanic and clastic rocks which are cut by gabbro; these are then intruded by numerous lamprophyre dikes of Cretaceous age. The age of the Leading Tickles dunite which intrudes the Ordovician Wild Bight Group has been the subject of a divergence. Both Jurassic and Ordovician ages have been suggested.

The lamprophyre dikes and the Leading Tickles dunite were sampled for a paleomagnetic study to define the age of magnetization and the tectonic implications of their emplacements.

Geology of the Notre Dame Bay Area

The Notre Dame Bay area is situated in the northeastern part of the central mobile belt. The area is underlain by Ordovician volcanic, pyroclastic, and sedimentary rocks. These rocks were involved in two orogenies, Taconic and Acadian, the second one producing the major structural trend (northeast folds and faults) (Harris 1973; Dean and Strong 1977). The youngest rocks of the area are the Budgell Harbour Gabbro of Jurassic age (Helwig *et al.* 1974) and some lamprophyre dikes of Jurassic to Cretaceous age (Strong and Harris 1974, Table 1).

In the sampling area (Fig. 2), four rock units can be recognized: the Wild Bight Group (Lower to Middle Ordovician, Penny's Brook Formation), the Exploits Group (Upper Ordovician – Lower Silurian, Point Leamington Formation), the Leading Tickles dunite (Ordovician or Jurassic?), and lamprophyre dikes (Jurassic–Cretaceous) (Harris 1973).

The Wild Bight Group consists of volcanic and

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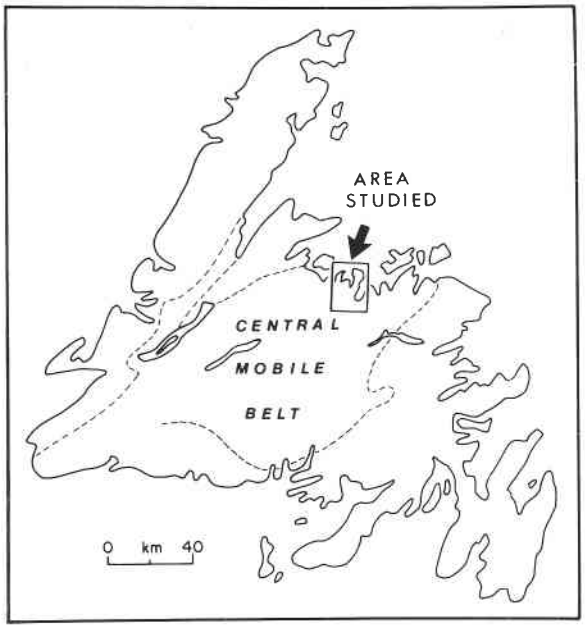


FIG. 1. Map of Newfoundland showing the area studied in the Central Mobile Belt and boundary of the other tectonic zones.

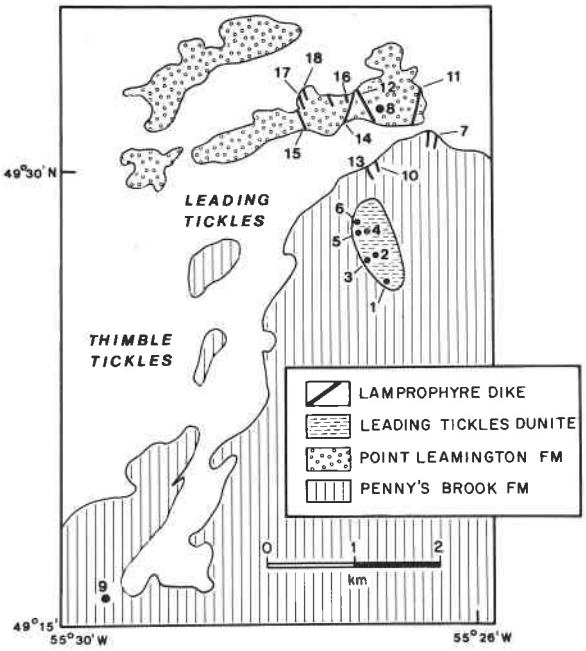


FIG. 2. General geological map of the Leading Tickles area (after Harris 1973). Numbered full dots show sampling sites in the Leading Tickles dunite. Numbers are associated with dikes sampled. Sites 8 and 9 yield no stable magnetization.

sedimentary rocks (Williams 1964). In the study area, red and green tuffs and cherts are predominant. The Exploits Group which includes the Point Leamington Formation, consists of greywacke and argillite (Williams 1962).

The Leading Tickles dunite is about 1 km long and about 0.5 km wide, and is well exposed along the road to Leading Tickles settlement. It can be best described as a differentiated diabase stock with an olivine rich zone. This intrusion was named by Helwig (1967) and was interpreted as Jurassic in age based on correlation with the Budgell Harbour Gabbro and related lamprophyres. However, using geochemical and petrographical work as a correlative factor, Harris (1973) suggested an Ordovician age for the Leading Tickles dunite.

The lamprophyres dikes were first described by Heyl (1935) as monchiquites and fourchites, although Helwig (1967) indicated that the most common are vogesite, spessartite, and kersantite. They range from minor veinlets to dikes 1 m in thickness, most of them being vertical to subvertical, and dominantly northeast trending. They cut Ordovician to Devonian rocks and have been dated as Jurassic to Cretaceous in age (Table 1).

Sampling and Techniques

Sample collections consisted of 123 cores (2.5 cm in diameter) from 18 sites, six from the Leading Tickles dunite, one from the Point Leamington Formation, one from the Wild Bight Group, and 10 from the lamprophyre dikes. From a core one, two, or three specimens (height 2.2 cm) were obtained and a total of 270 specimens were studied.

Two cleaning treatments were used; alternating field (AF) and thermal. The experimental work was carried out using equipment operating in residual fields 2 nT (nanotesla) (Roy *et al.* 1969, 1972, 1973) and a commercial Schonstedt thermal unit (TSM-1). All measurements were performed using a digital Schonstedt spinner magnetometer (DSM-1). For all thermally treated specimens steps of 100°C from 100–500°C and thereafter 550, 560, and 570°C were used. Some specimens were subjected to temperatures up to 670°C. All AF treated specimens were subjected to fields of 5, 10, 15, 20, 22.5, 25, 27.5, 30, and 50 mT. Some specimens were subjected to treatments up to 125 mT.

Results

Lamprophyre Dikes

From the vicinity of the Leading Tickles settlement, 68 cores (150 specimens) were obtained from 10 different dikes (one site per dike), yielding a single magnetization related to a Lower Cretaceous age. The single normal polarity NRM (natural remanent magnetization) site-mean directions are fairly well grouped around 331°, +73° (Fig. 3, Table

TABLE 1. Radiometric ages from Notre Dame Bay area

Rock type	Age (Ma)	Method	Reference
Lamprophyre dike	115 ± 20	K/Ar	Wanless *et al.* (1967)
	129 ± 7	K/Ar	Wanless *et al.* (1967)
	144 ± 12	K/Ar	Wanless *et al.* (1965)
Budgell Harbour stock	155 ± 7	K/Ar	*In* Strong and Harris (1974)
	135 ± 8	K/Ar	Helwig *et al.* (1974)
	139 ± 9	K/Ar	Helwig *et al.* (1974)
Leading Tickles dunite	Cretaceous–Jurassic	Petrographic affinity	Helwig (1967)
	Ordovician	Petrochemical affinity	Harris (1973)

2). The intensities of the different specimens vary from 700 × 10 to 1500 × 10^{-3} A/m.

Upon thermal treatment, most of the direction changes occur below 100°C (Fig. 4*a*). The direction changes may be attributable to the removal of a viscous remanence, or partial thermoremanent magnetization (PTRM). A portion (≈30%) of all remanence has unblocking temperatures (T_{UB}) between 100 and 500°C; small direction changes occur, perhaps reflecting the secular variation of the field during cooling of the intrusion. The T_{UB} of most of the remanence is confined to a very short range (500–550°C, Fig. 4). After treatment at 550°C and higher, the directions become random and no thermally stable remanence appears to survive.

Alternating field treatment also shows that the NRM is composed of two remanences. One of the remanences has remanent coercive force (rcf) below 5 mT, as shown by the direction change (Fig. 5*a*). The predominant remanence has rcf distributed between 5 and 25 mT; small direction changes are observed in this interval (Fig. 5*a*). The site-mean directions cluster about 328°, +61°, in close agreement with the thermally obtained mean. The within-site k are much greater after than before treatment indicating that the scatter of NRM directions is mostly attributable to the low rcf remanence (Table 2). From a comparison of thermal and AF results it would appear that the remanence with T_{UB} below 100°C corresponds to the remanence with rcf of 5 mT. The remanence with T_{UB} between 100 and 500–550°C appears to have rcf between 5–10 and 25 mT (Figs. 4, 5).

Site-mean directions were calculated for NRM, thermal, AF, and combined results (Table 2). The k for the NRM, thermally treated, and AF treated site means are 54, 89, and 131 respectively. The combined AF and thermal results yield a k value of 180, an increase of 333% over the NRM's k (54). The paleopole obtained falls at 212°E, 67°N with an α_{95} of 3.6°. It should be noted that a collection of lamprophyre dikes from the Fortune Harbour area are presently under study (Deutsch and Rao 1977).

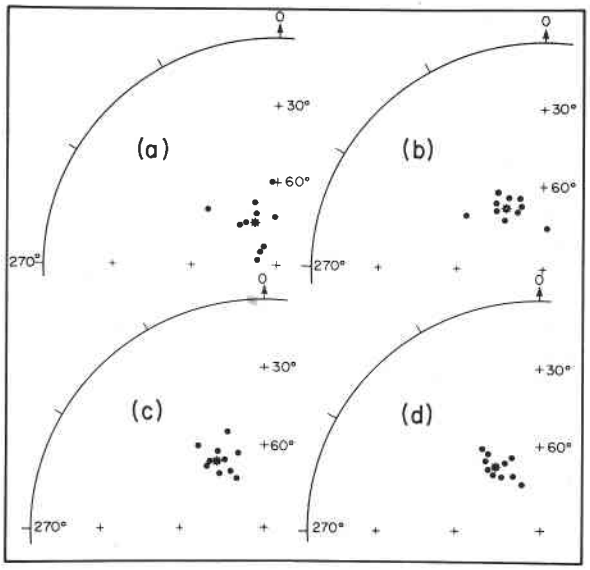


FIG. 3. Shows stereonet projection (equal area) of lamprophyre dikes before and after cleaning. Full dot is individual site mean (positive inclination). Star symbol represents the site-mean direction. (*a*) NRM site-mean distribution. (*b*) Thermal site-mean distribution. (*c*) AF site-mean distribution. (*d*) AF and thermal (*combined*) site-mean distribution.

Leading Tickles Dunite

The Leading Tickles dunite, as previously noted, is the subject of a divergence with regard to its age. It has been assumed that the intrusion was emplaced during the Cretaceous igneous event (Helwig 1967). However, it will become apparent that the dunite body is probably of Ordovician age (Harris 1973).

The site mean NRM's from the six sites sampled are plotted in Fig. 6. Unlike those of the lam-

TABLE 2. Site-mean values for Leading Tickles lamprophyre dikes

	NRM			Thermal			AF			Combined				
Site	n	$D°, I°$	k	N	$D°, I°$	k	N	$D°, I°$	k	N	$D°, I°$	k	Lat. (N), Long. (E)	$\alpha_{95}°$
7	09	322, 82	03	03	006, 75	90	08	331, 70	42	11	338, 72	43	75, 251	7.0
10	06	340, 66	12	04	329, 57	119	04	317, 59	169	08	323, 58	126	62, 206	7.2
11	08	356, 72	15	04	336, 67	166	04	321, 65	118	08	323, 65	154	65, 222	4.5
12	08	323, 71	18	04	342, 63	20	04	329, 58	354	08	332, 62	40	70, 207	8.8
13	06	309, 58	14	06	322, 63	145	06	321, 60	113	12	321, 61	136	62, 215	3.7
14	08	323, 71	10	04	324, 61	85	03	322, 52	1515	07	325, 56	160	62, 200	4.8
15	05	356, 60	05	03	333, 61	134	03	342, 62	343	06	338, 61	199	73, 199	4.8
16	05	308, 83	04	02	341, 66	210	03	330, 68	39	05	334, 68	56	73, 129	10.3
17	05	337, 70	11	03	305, 56	58	04	339, 53	155	07	326, 55	38	61, 198	9.9
18	08	283, 84	02	02	322, 68	14	03	329, 62	298	05	323, 64	42	65, 219	12.0
Mean	10	331, 73	54	10	330, 64	89	10	328, 61	131	10	328, 62	180	67, 212	3.6

NOTES: n = number of cores; N = number of specimens; $D°, I°$ = declination and inclination; k = best precision estimates (Fisher 1953); and α_{95} = radius of circle of confidence ($P = 0.05$).

TABLE 3. Site-mean values for the Leading Tickles dunite

	NRM				Thermal				AF				Combined				
Site	n	$D°$	$I°$	k	N	$D°$	$I°$	k	N	$D°$	$I°$	k	N	$D°$	$I°$	k	α_{95}
1	6	310	+18	01	4	356	+59	33	2	325	+70	20	6	357	+65	19	15.9
2	7	039	−70	03	3	181	+64	33	6	109	+74	19	9	142	+74	15	13.5
3	7	050	−55	10	5	242	+42	07	7	137	+70	18	12	215	+64	07	18.3
4	7	243	+28	05	3	246	+24	90	6	195	+36	1316	9	213	+35	13	15.1
5	7	228	−15	11	4	211	+14	28	10	195	+34	131	14	199	+29	23	8.5
6	8	029	+59	02	3	005	+67	66	3	001	+68	61	6	003	+67	78	7.6

NOTES: Sites mean (1 + 6): 000°, +66°; α_{95} = 6.8; 89°N, 129°E. Sites mean (2 + 3 + 4 + 5): 201°, +52°; α_{95} = 31°; 107°E, 5°N.
n = number of cores; N = number of specimens; $D°, I°$ = declination and inclination; k = best precision estimate (Fisher 1953); and α_{95} = radius of circle of confidence ($P = 0.05$).

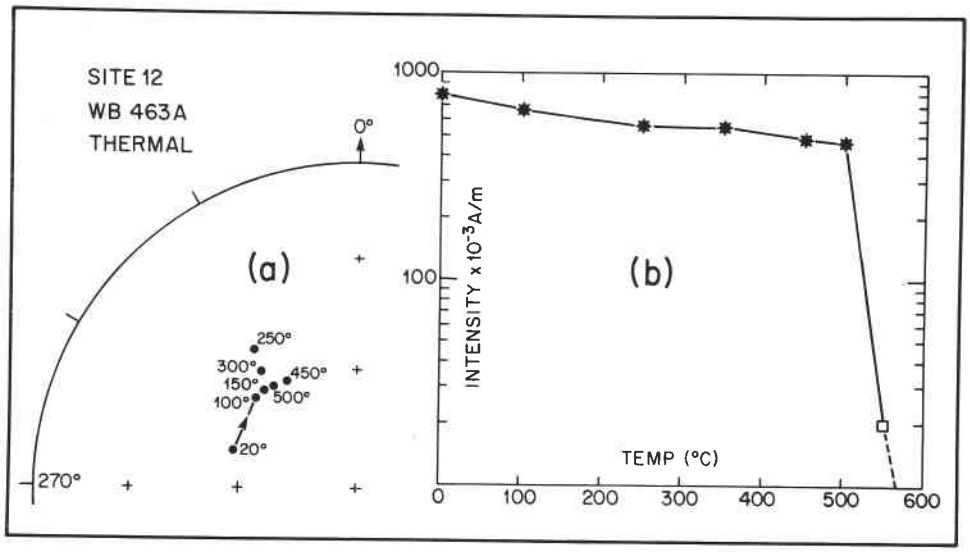


FIG. 4. Characteristic thermal stepwise demagnetization of lamprophyre dikes sample. (*a*) Equal area stereographic projection showing changes of direction during treatment. Full dot represents positive inclination (normal). From 20–100°C a soft component is removed. The direction is stable up to 500°C. After 500°C the directions became random. (*b*) Intensity decay curve for the same specimen shown in (*a*). From 20–500°C the specimen lost 45% of its initial intensity. At 550°C less than 5% of the total initial intensity remains. The dotted line signifies that other measurements were performed but were found to be unstable.

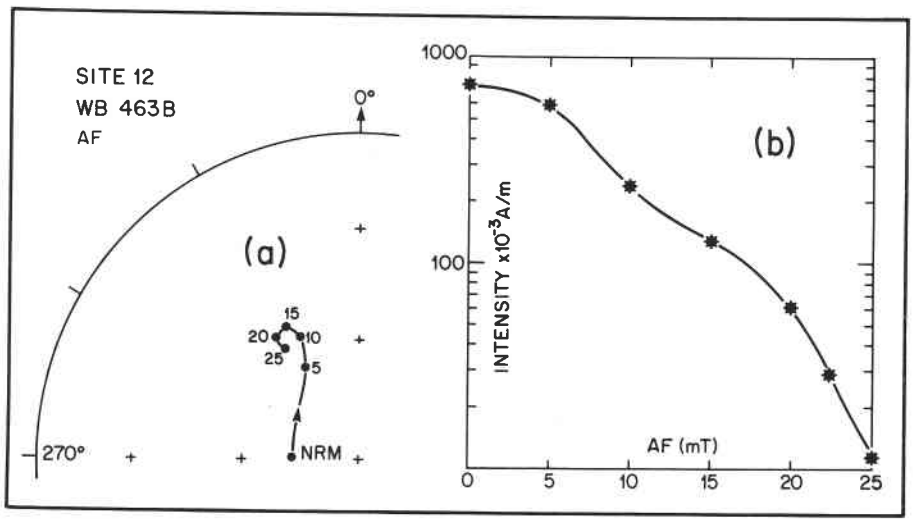


FIG. 5. Alternating field stepwise demagnetization on lamprophyre dikes samples. (*a*) Changes in direction during AF treatment. From 0–5 mT a small change in direction occurs; after 5 mT the direction is stable up to 25 mT. Thereafter the directions are randomly distributed. (*b*) Intensity decay curve for the same specimen. After 20 mT less than 10% of the initial intensity remains. After 25 mT no remanence of any stability could be detected.

prophyre dikes NRM (Fig. 3), the directions are quite scattered; 42 cores (100 specimens) were subjected to thermal (up to 675°C) or alternating field (up to 130 mT) treatment.

After demagnetization, the mean direction for two sites (1 and 6, Fig. 6) moves toward a high inclination direction which is similar to the mean direction observed in the lamprophyre dikes (Tables 2, 3). The directions at the other four sites (2–5) move towards the southwest quadrant, now with positive polarity (Table 3, Fig. 6). Although the mean directions of the four sites are quite scat-

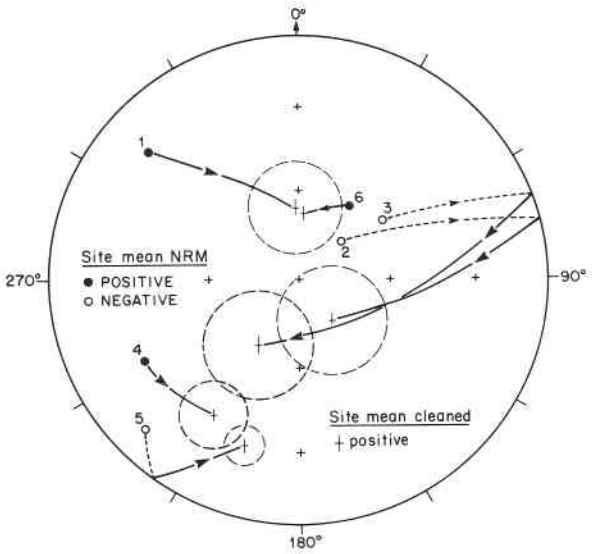


FIG. 6. Site-mean NRM (full dots) and site-mean cleaned direction (crosses) for the six sites of the Leading Tickles dunite. Full (open) dots are positive (negative) inclination, crosses are positive inclination. Broken lines are related α_{95} of each site mean. It is clearly seen that after demagnetization sites 1 and 6 (thermal and AF) group in a direction which is near the direction obtained for the lamprophyre dikes site-mean direction. Sites 2, 3, 4, and 5 tend to cluster into a direction in the third quadrant with a low positive inclination. This direction is probably most indicative of the direction at the time of intrusion and is interpreted to be of Lower Paleozoic age.

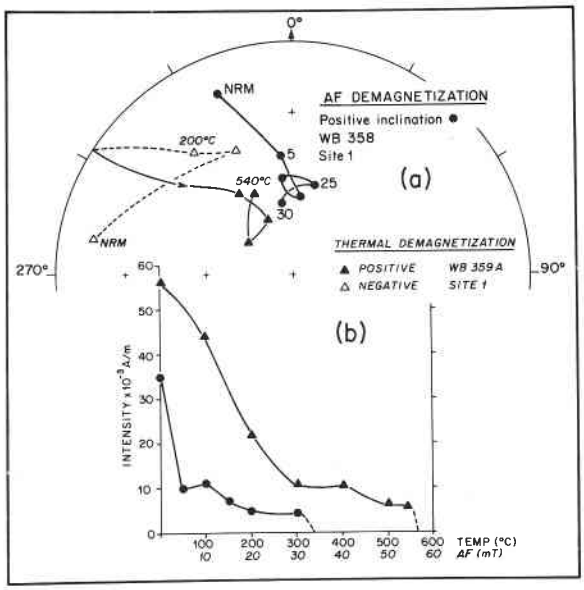


FIG. 7. Characteristic thermal and AF stepwise demagnetization from site 1 and 6 interpreted as a probable Lower Cretaceous overprint. (*a*) Stereonet projection of direction changes during thermal and AF treatment. In both cases a soft component is removed after 300°C and 5 mT. Dotted line means that further treatments were done after 540°C and 30 mT, but the results fail to indicate remanence of any stability. (*b*) Intensity decay curve for both AF and thermal treatment. The thermal curve shows a large drop up to 300°C and then stabilizes in a plateau until 540°C. A similar pattern is seen in the direction stability. On AF treatment, a large drop at 5 mT and a constant decay up to 30 mT is seen.

tered their mean indicates that at least part of the remanence was acquired at the time of intrusion, probably in Ordovician time, whereas the remanence at sites 1 and 6 was probably acquired during the Cretaceous igneous activity.

Sites 1 and 6

Natural remanence magnetization intensity varies from 30×10^{-3} to 100×10^{-3} A/m. Upon thermal and AF demagnetization the directions tend to cluster about 000°, +66° (Table 3, Fig. 7). This direction is reached at about 300°C. From 300–540°C small direction changes occur. This might be attributable to a superimposed magnetization or perhaps to the effect of secular variation. No remanence of any stability could be detected after treatment at 540°C and higher temperatures.

Alternating field demagnetization also shows similar changes. The direction (000°, +66°) is reached at 10 mT and remains approximately in that position until 30 mT. Small changes in direction similar to those observed thermally occur in that range. After 30 mT the directions are randomly distributed. The site-mean pole falls at 129°E, 89°N (Table 3).

Sites 2–5

Comparison of site mean directions before and after treatment clearly shows that a large shift in direction occurs in three of the four sites (Fig. 6), the inclination changing from negative to positive in both AF and thermally treated samples. Natural remanent magnetization intensities vary from 300×10^{-3} to 1000×10^{-3} A/m.

During AF treatment, the direction changes occur below 10 mT (Fig. 8). From 10–50 mT little change in direction occurs for sites 4 and 5 and in most cases the remanence resists fields of 75 mT; the mean direction is 206°, +32°. At sites 2 and 3 small changes in direction occur between 10 and 50 mT (Fig. 8) and the highest rcf is 50 mT. The directions are not as tightly grouped as for sites 4 and 5 and the mean directions are different. It is interpreted that at sites 2 and 3 the direction (≈200°, +50) does not represent a single remanence but is the result of remanences of either contemporaneous ages (dual polarity) or distinctly different ages (Ordovician and Cretaceous?). Experimental work indicates that the former is more probable.

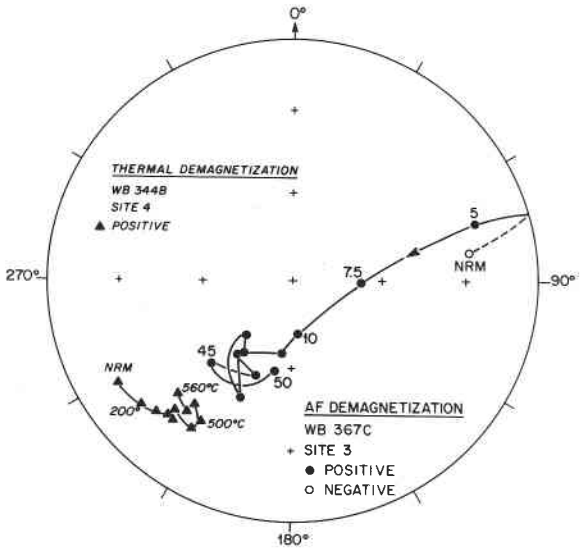


FIG. 8. Characteristic thermal and AF stepwise demagnetizations from sites 3 and 4, which are interpreted to represent the original magnetization, are shown. It can be seen in the AF demagnetization that a soft component is removed after 10 mT. The direction is stable up to 50 mT. On thermal demagnetization a very small direction change occurs between 20 and 200°C, and the direction is stable up to 560°C. Samples shown came from two different sites (3 and 4) having quite different NRM. Although the grouping is scattered (Table 3) these results are interpreted as showing a Lower Paleozoic age for the Leading Tickles dunite.

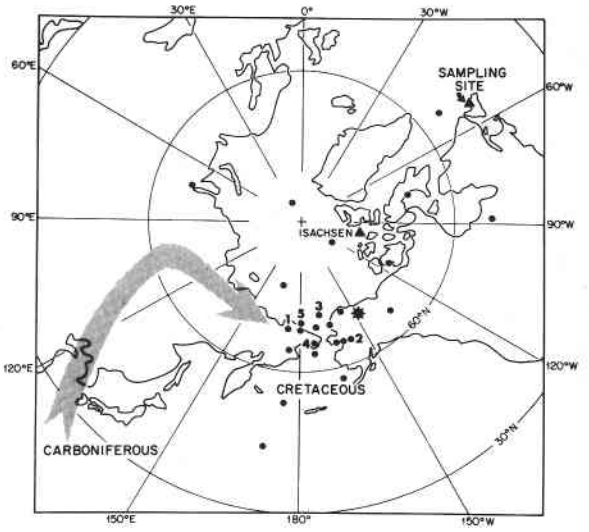


FIG. 9. This figure shows the pole path from Carboniferous to Cretaceous for Laurentia. All Cretaceous poles are plotted as full dot. Pole numbers from 1–5 are: (1) Mont Megantic: 172°E, 69°N (Larochelle 1968); (2) Monteregian Hills 203°E, 65°N (Larochelle 1968); (3) Monteregian Hills 190°E, 71°N (Larochelle 1968); (4) Mount Ascutney 187°E, 64°N (Opdyke and Wensink 1966); and (5) Isachsen diabase 180°E, 69°N (Larochelle and Black 1963). The star pole is the pole from this study, the Notre Dame Bay lamprophyre dikes ($\alpha_{95} = 3.6°$). Triangles show the location of the area studied and the Isachsen diabase dikes.

Upon thermal demagnetization sites 2–5 give stable directions from 300–560°C, with only small changes in directions over this range, as in the AF treatment (Fig. 8). After heating above 570°C the directions are random.

The fact that under both thermal and AF treatments the remanences behave similarly at all four sites indicates that the remanences have similar magnetic characteristics and hence similar origins and ages. This interpretation favours the dual polarity option as an explanation for the dispersion found at sites 2 and 3. This interpretation is supported by the fact that the NRM's are negatively inclined and generally opposite to the directions obtained after treatment. If the two remanences had been acquired at distinctly different times (and presumably under different conditions), one would expect the direction of magnetization to move toward the Ordovician or Cretaceous direction upon thermal or AF treatment. Whatever the cause, the large within-site dispersions encountered at sites 2 and 3 render the site-mean direction highly suspect as a record of the field direction at any time. The mean of sites 4 and 5 then is considered a better *estimate* of the direction of the field at the time of intrusion, and it yields a pole at 098°E, 19°N.

The sites from Point Leamington and the Wild Bight Group yield no stable magnetization.

Discussion

Lamprophyre Dikes

The single normal polarity direction (328°, +62°) which has been obtained from 10 different lamprophyre dikes in the Notre Dame Bay area is associated with a Lower Cretaceous age. The paleopole falls at 212°E, 67°N (Table 2) near the Cretaceous pole cluster for North America (Fig. 9, Irving *et al.* 1976). The pole path from Jurassic to Cretaceous for North America follows an eastward direction along the present day latitude of 70° (Steiner and Helsley 1975). The paleopole derived from lamprophyre dikes is in good agreement with the radiometric data (Table 1) and the probable age is Lower Cretaceous. These dikes correspond to the youngest igneous activity event (interval 100–130 Ma) in eastern North America and are probably related to the White Mountain series, the Monteregian Hills intrusion, and the Mafic dikes intrusions of eastern North America (Foland and Faull 1977; McHone 1978; Harris 1973).

The predominance of *normal* polarity in the lamprophyre dikes is also an indication of a Lower

Cretaceous age; it can be used to demonstrate the spatial relationship of different intrusions. The Cretaceous polarity time scale is characterized by a *normal* polarity interval (Larson and Pitman 1972; McElhinny and Burek 1971; Irving and Pullaiah 1976). The Jurassic–Cretaceous (110–150 Ma) and the Cretaceous–Tertiary (50–81 Ma) boundary are characterized by a *mixed* polarity interval. It is then probable that the lamprophyre dikes were intruded during the Cretaceous *normal* (81–110 Ma) polarity interval. It should be noted that an alternative explanation would be that the dikes were intruded in a normal polarity time within a mixed polarity interval. This explanation implies that the time of intrusion and the cooling rate of these dikes were rapid and it would also imply that no chemically acquired remanences are present. Since this explanation is more demanding, the previous explanation is favoured. So, if the dikes were intruded in the normal polarity interval, then the Mont Megantic (172°E, 69°N) and the Monteregian Hills paleopoles (203°E, 65°N and 190°E, 71°N, Larochelle 1968) are probably slightly older than the Notre Dame Bay lamprophyre dikes. Their results show *mixed* polarities, associating the intrusions with the Jurassic–Cretaceous *mixed* polarity interval. The Isachsen diabase dikes from the Canadian Arctic Archipelago (Larochelle and Black 1963), paleopoles (180°E, 69°N), show only *normal* polarity, and, therefore, are more likely to be of the same age as the Notre Dame Bay dikes. Radiometric studies of the Isachsen diabase showed a Lower Cretaceous age (Larochelle *et al.* 1965).

Leading Tickles Dunite

Although the paleopole (098°E, 19°N) obtained from the Leading Tickles dunite is only a rough estimate of the pole at the time of intrusion its location implies a Lower Paleozoic age for this intrusion (Lapointe 1979). This conclusion supports Harris' (1973) idea based on geochemical considerations that the intrusion is related to the Wild Bight Group (Ordovician) and not to the lamprophyre dikes (Cretaceous).

The large dispersion of direction found at sites 2 and 3 could be attributed to one or more of the following causes. It could be due to an overprint acquired during the Cretaceous igneous activity (Table 3). A preferred explanation is of geological origin; the magnetization found in this dunite is probably a chemical remanent magnetization (CRM) due to serpentinization of the body, occurring as part of the initial formation process (Harris 1973; Saad 1969). Similar effects are seen in the Franciscan ultramafic rocks (Saad 1969) where a CRM is related to serpentinization. The acquisition time of a CRM could easily span field-reversal time scales. The result would be to produce nearly opposite magnetizations of comparable characteristics. Because the paleopole of the Leading Tickles dunite is in agreement with the other Lower Paleozoic poles (Lapointe 1979), this rock unit is considered to be of Ordovician rather than Jurassic age. However, since the direction of the remanence cannot be defined accurately, the pole itself should not be used as a key reference pole.

Conclusions

It is of interest to note the fact that the Newfoundland lamprophyre dikes are probably the youngest igneous event recorded in eastern Canada. One suspects that the origin of these dikes is related to the opening of the North Atlantic Ocean (Laughton 1975; McHone 1978). This has been suggested on geological grounds by Currie (1970). What happens when the North Atlantic Ocean opens?

It is believed "that the opening of the ocean would create a compressional regime between northern Eurasia and the Arctic–Alaska plate. Consequently the Arctic–Alaska plate rotated away from Laurentia and thereby opened the Amerasia basin" (Sweeney *et al.* 1978, p. 106). One effect of such tectonic regimes is the intrusion of dikes in areas of weakness subjected to these regimes. If the hypothesis of Sweeney *et al.* (1978) and Laughton (1975) holds, i.e., that the rotation of the Arctic–Alaska plate and the North Atlantic Ocean opening were part of the same regime, then it is possible that the Isachsen diabase and the Newfoundland lamprophyre dikes are the signature of the same tectonic regime.

If this is true, then these two intrusions should have the same age, the same paleopole, the same polarity, and be intruded in regions tectonically affected. In both cases the radiometric ages (Larochelle *et al.* 1965; Wanless *et al.* 1967) corroborate a Lower Cretaceous age. Similarly, both paleopoles (Isachsen 180°E, 69°N and Nfld. 212°E, 67°N) are in agreement, within the resolution power of paleomagnetism. In the two studies (Larochelle and Black 1963; and this paper) one single *normal* polarity was found.

Although these relationships could be coincidental, the Sweeney *et al.* (1978) hypothesis seems to provide a logical reference frame in which the present data fit easily. The implication that the Notre Dame Bay dikes are the signature of the opening of the North Atlantic Ocean is that the timing of the opening is probably Lower Cretace-

ous in age. Consequently, it also implies that the Arctic–Alaska plate has been formed as a result of rotation away from Laurentia, and that this rotation is part of the same tectonic regime as the opening of the North Atlantic Ocean.

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